

Grade 7
Unit 9
Lesson 4 Practice

Name Answer Key
Date _____
Period _____

Find the sum of each expression by combining like terms.

1. $(0.6x + 0.2) + (0.9x - 0.3) + (2.8x - 6.6)$

$$0.6x + 0.9x + 2.8x + 0.2 - 0.3 - 6.6$$

$$4.3x - 6.7$$

2. $\left(\frac{1}{5} - 6y\right) - \left(3y - \frac{4}{5}\right) - \left(y + \frac{2}{5}\right)$

$$\frac{1}{5} - 6y - 3y + \frac{4}{5} - y - \frac{2}{5}$$

$$-6y - 3y - y + \frac{1}{5} + \frac{4}{5} - \frac{2}{5}$$

$$-10y + \frac{3}{5}$$

3. $(0.5z) - (-0.4z + 8) + (0.3z + 8)$

$$0.5z + 0.4z - 8 + 0.3z + 8$$

$$1.2z$$

4. $\left(5\frac{1}{2}y + \frac{3}{5}\right) + \left(1\frac{3}{5} - 3\frac{1}{4}y\right) - \left(6y - \frac{2}{5}\right)$

$$5.5y + 0.6 + 1.6 - 3.25y - 6y + 0.4$$

$$5.5y - 3.25y - 6y + 0.6 + 1.6 + 0.4$$

$$-3.75y + 2.6$$

$$-3\frac{3}{4}y + 2\frac{3}{5}$$

5. $\left(2 - \frac{7}{9}q\right) - \left(-3 - \frac{2}{9}q\right) - \left(-5 - \frac{5}{9}q\right)$

$$2 - \frac{7}{9}q + 3 + \frac{2}{9}q + 5 + \frac{5}{9}q$$

$$-\frac{7}{9}q + \frac{2}{9}q + \frac{5}{9}q + 2 + 3 + 5$$

$$10$$

Simplify each of the expressions.

6. $(7.6 - 4.9x) - (2.8x - 3.7) - (9.8x - 4.2)$

$$7.6 - 4.9x - 2.8x + 3.7 - 9.8x + 4.2$$

$$-4.9x - 2.8x - 9.8x + 7.6 + 3.7 + 4.2$$

$$-17.5x + 15.5$$

7. $\left(\frac{1}{8}x - \frac{3}{4}\right) + \left(\frac{3}{4}x + \frac{13}{16}\right) + \left(\frac{3}{8}x - \frac{1}{4}\right)$

$$\frac{1}{8}x - \frac{12}{16} + \frac{6}{8}x + \frac{13}{16} + \frac{3}{8}x - \frac{4}{16}$$

$$\frac{1}{8}x + \frac{6}{8}x + \frac{3}{8}x - \frac{12}{16} + \frac{13}{16} - \frac{4}{16}$$

$$\frac{10}{8}x - \frac{3}{16}$$

$$\frac{5}{4}x - \frac{3}{16}$$

$$1\frac{1}{4}x - \frac{3}{16}$$

8. $(2.8y + 8) - (2.7 + 3y) + (-8.7 + 5y)$

$$2.8y + 8 - 2.7 - 3y - 8.7 + 5y$$

$$2.8y - 3y + 5y + 8 - 2.7 - 8.7$$

$$4.8y - 3.4$$

9. $\left(2.8 + \frac{4}{9}r\right) + (-6.8 + r) - \left(-4.7 + \frac{7}{9}r\right)$

$$2.8 + \frac{4}{9}r - 6.8 + r + 4.7 - \frac{7}{9}r$$

$$\frac{4}{9}r + r - \frac{7}{9}r + 2.8 - 6.8 + 4.7$$

$$\frac{6}{9}r + 0.7$$

$$\frac{2}{3}r + 0.7$$

10. $\left(\frac{5}{6} + \frac{3}{7}n\right) - \left(-\frac{1}{6} + n\right) - \left(2\frac{5}{6} + 7\frac{2}{7}n\right)$

$$\frac{5}{6} + \frac{3}{7}n + \frac{1}{6} - n - 2\frac{5}{6} - 7\frac{2}{7}n$$

$$\frac{3}{7}n - n - 7\frac{2}{7}n + \frac{5}{6} + \frac{1}{6} - 2\frac{5}{6}$$

$$\frac{3}{7} - \frac{7}{7} - \frac{51}{7} = -\frac{55}{7} = -7\frac{6}{7}$$

$$\frac{5}{6} + \frac{1}{6} - \frac{17}{6} = -\frac{11}{6} = -1\frac{5}{6}$$

$$-7\frac{6}{7}n - 1\frac{5}{6}$$